

A bisection identity between OEIS A155543 and OEIS A090129

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Abstract

OEIS A155543 carries a conjecture that its n -th term equals the $(2n + 2)$ -nd term of OEIS A090129, the multiplicative order of 3 modulo 2^n . It is true. Both entries state closed forms as fact — A090129's is the order computation 2^{n-2} for $n \geq 3$, recorded there with a proof, and A155543's is a rational generating function — and the two match term for term once the even index $2n + 2$ is substituted. The identity holds at every index, including the one the comment excludes.

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1 The two sequences and the conjecture

OEIS A155543 is “ $a(n) = 2 \cdot A081294(n)$.”. It has offset 0 and begins

2, 4, 16, 64, 256, 1024, 4096, ...

OEIS A090129 is “Smallest exponent such that $-1 + 3^{a(n)}$ is divisible by 2^n .”. It has offset 1 and begins

1, 2, 2, 4, 8, 16, 32, 64, ...

The entry A155543 carries the following comment:

Conjecture: $a(n) = A090129(2 \cdot n + 2)$. — *the entry (unattributed)*, undated

As of the “Last modified” line on the live entry (Jun 30 2026 22:40:50, revision 16) the statement is still recorded as unproved, and nothing on either entry records it as settled either way.

Written out, the statement to be proved is

$$a(n) = b(2n + 2), \tag{1}$$

where a is A155543 and b is A090129. A090129 has offset 1, so $b(2n + 2)$ is defined for every $n \geq 0$.

2 What each entry states as fact

A090129 is defined as the smallest exponent $b(n)$ with $2^n \mid 3^{b(n)} - 1$, that is, the multiplicative order of 3 modulo 2^n . The entry states, not as a conjecture,

$a(n) = 2^{n-2}$ if $n \geq 3$, 1 for $n=1$ and 2 for $n=2$ (see the order comment above).

that is,

$$b(1) = 1, \quad b(2) = 2, \quad b(n) = 2^{n-2} \quad (n \geq 3). \tag{2}$$

The entry also carries the reason, in a comment of W. Lang: for $n \geq 3$ the order of 3 modulo 2^n is 2^{n-2} , with a reference for the argument. So that nothing here rests on a reference, that fact is proved from scratch in Section 4; (2) is then not an assumption at all.

A155543 states, not as a conjecture,

G.f.: $2^*(1-2^*x)/(1-4^*x)$.

Expanding, $\frac{2(1-2x)}{1-4x} = 2 \sum_{k \geq 0} 4^k x^k - 4 \sum_{k \geq 0} 4^k x^{k+1} = 2 + \sum_{k \geq 1} (2 \cdot 4^k - 4 \cdot 4^{k-1}) x^k$, and $2 \cdot 4^k - 4^k = 4^k$, so

$$a(0) = 2, \quad a(n) = 4^n \quad (n \geq 1). \quad (3)$$

Each was checked against its own entry's DATA at every published index before use: (3) against all 24 terms of A155543 and (2) against all 36 terms of A090129.

3 The order of 3 modulo a power of 2

Write $v_2(m)$ for the exponent of 2 in the integer $m \neq 0$.

Lemma 1. $v_2(3^{2^k} - 1) = k + 2$ for every $k \geq 1$.

Proof. For $k = 1$, $3^2 - 1 = 8$ and $v_2(8) = 3$. Suppose the claim for some $k \geq 1$. Factor

$$3^{2^{k+1}} - 1 = (3^{2^k} - 1)(3^{2^k} + 1).$$

Since $k \geq 1$, the inductive hypothesis gives $8 \mid 3^{2^k} - 1$, so $3^{2^k} \equiv 1 \pmod{8}$ and hence $3^{2^k} + 1 \equiv 2 \pmod{8}$, giving $v_2(3^{2^k} + 1) = 1$. Therefore $v_2(3^{2^{k+1}} - 1) = (k + 2) + 1 = (k + 1) + 2$. $\square$

Proposition 2. For $n \geq 3$ the multiplicative order of 3 modulo 2^n is 2^{n-2} ; it is 1 for $n = 1$ and 2 for $n = 2$. That is, (2) holds.

Proof. The group $(\mathbb{Z}/2^n\mathbb{Z})^\times$ has order 2^{n-1} , so the order of 3 divides 2^{n-1} and is therefore 2^k for some $k \geq 0$; it is the least such k with $2^n \mid 3^{2^k} - 1$, that is, with $v_2(3^{2^k} - 1) \geq n$.

Let $n \geq 3$. By Lemma 1 the condition reads $k + 2 \geq n$ for $k \geq 1$, and for $k = 0$ it reads $v_2(2) = 1 \geq n$, which fails. The least admissible k is thus $n - 2$, which is ≥ 1 exactly because $n \geq 3$. Hence the order is 2^{n-2} .

For $n = 1$, $3 \equiv 1 \pmod{2}$ and the order is 1. For $n = 2$, $3 \equiv 3 \pmod{4}$ and $3^2 = 9 \equiv 1$, so the order is 2. $\square$

4 The proof

Theorem 3. For every integer $n \geq 0$, $a(n) = b(2n + 2)$, where a is OEIS A155543 and b is OEIS A090129. This is the statement of Section 1.

Proof. Take $n \geq 1$. Then $2n + 2 \geq 4 \geq 3$, so Proposition 2 gives

$$b(2n + 2) = 2^{(2n+2)-2} = 2^{2n} = 4^n,$$

which is $a(n)$ by (3).

For $n = 0$ the index is $2n + 2 = 2$, and Proposition 2 gives $b(2) = 2$, while (3) gives $a(0) = 2$. The two agree, so the identity holds at $n = 0$ as well. $\square$

Corollary 4. For every $n \geq 1$ the multiplicative order of 3 modulo 2^{2n+2} is 4^n .

Proof. Immediate from the theorem and the definition of A090129, together with (3). $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the generating function of A155543 was expanded in exact rational arithmetic and compared against every one of the 24 published terms; the values agree exactly. A generating function that reproduced only some of them would not have been used.

Second, formula (2) was evaluated and compared against every one of the 36 published terms of A090129; the values agree exactly. Independently, the order of 3 modulo 2^n was computed directly by repeated squaring for every n in that range and matched the same values, so the closed form was not taken on trust; and Lemma 1 was checked by computing $v_2(3^{2^k} - 1)$ exactly for $k \leq 12$, giving 3, 4, 5, $\dots$, 14 as the lemma requires.

Third, the identity (1) was checked directly on the published terms, without either formula: for every n from 0 to 17 — the range for which both $a(n)$ and $b(2n + 2)$ are published — the two integers agree.

References

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- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011.